Lecture 2 - Graphs and Dynamic Programming

Course: Algorithm Design and Analysis (ADA101)
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Summary

Short lecture material for graph representation, breadth-first search, depth-first search, and dynamic programming principles.

Graph representation

- An adjacency list stores each vertex with the vertices connected to it.
- An adjacency matrix stores connections in a two-dimensional table.
- Lists are usually better for sparse graphs, while matrices are useful for fast edge lookup.

Search methods

- Breadth-first search explores neighbors level by level and is useful for shortest paths in unweighted graphs.
- Depth-first search follows a path deeply before backtracking and is useful for components, cycles, and topological ordering.

Dynamic programming

- Dynamic programming solves a problem by reusing answers to smaller overlapping subproblems.
- The technique usually requires a recurrence relation and a table or cache.

Quick reference

Topic	When to use	Main cost
BFS	Unweighted shortest path	O(V + E)
DFS	Reachability and cycle checks	O(V + E)
DP	Overlapping subproblems	Depends on state count